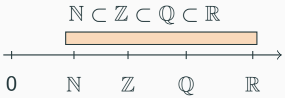
Math Bootcamp

Day 1 — Foundations (Sections 1.1–1.4)

Math Camp 2026

1.1 Number systems — names and definitions

Symbol	English name	What it is
$\mathbb{N}$	natural numbers	Non-negative integers for **counting**
$\mathbb{Z}$	integers	$\ldots, -2, -1, 0, 1, 2, \ldots$
$\mathbb{Q}$	rational numbers	Fractions p/q $(q \neq 0)$
$\mathbb{R}$	real numbers	All points on the number line
$\mathbb{R}^n$	n-dimensional real space	Lists / vectors of length n



1.1 Symbols and vectors — names and definitions

Common symbols	$\in$	"is an element of"	$\forall$	"for all"
	$\exists$	"there exists"	$\Rightarrow$	"implies"
			$\Leftrightarrow$	"if and only if" (iff)

Scalar vs. vector

- **Scalar:** one number, e.g. income $w \in \mathbb{R}$
- **Vector:** a list of numbers $\mathbf{x} = (x_1, \ldots, x_n) \in \mathbb{R}^n$
- **Dot product:** $\mathbf{p} \cdot \mathbf{x} = \sum_{i=1}^{n} p_i x_i$

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$\to \mathbb{Q}$ e.g. $8.50 = 17/2$ dollars

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3. Order as a vector: (HK\$ spent on drink, # snacks) $=$ (8.50, 2)

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$\to\ (8.50, 2) \in \mathbb{Q} \times \mathbb{N}$

1.2 Logic — names and definitions

English	Symbol	Meaning
P **only if** Q	$P \Rightarrow Q$	Q is **necessary** for P (without Q, no P)
P **if** Q	$Q \Rightarrow P$	Q is **sufficient** for P (Q alone is enough)
P **if and only if** Q	$P \Leftrightarrow Q$	Q is necessary **and** sufficient
Contrapositive	$\neg Q \Rightarrow \neg P$	same truth as $P \Rightarrow Q$: if Q fails, P must fail

1.1 — IKEA run Together

First week in HK: you go to IKEA and spot three plush toys.

Otter HK$179.9

Hedgehog HK$79.9

BLÅHAJ HK$179.9

You have **HK$500** pocket money ($B = 500$).

Basket $\mathbf{x} \in \mathbb{N}^3$: # otters, # hedgehogs, # sharks. E.g. $(1, 1, 0)$ = one otter, one hedgehog, no shark.
Price vector $\mathbf{p} = (179.9, 79.9, 179.9) \in \mathbb{Q}^3$. Budget set: $\mathcal{B} = \{\mathbf{x} \in \mathbb{N}^3 : \mathbf{p} \cdot \mathbf{x} \leq 500\}$.

$\mathbf{p} = (179.9,\ 79.9,\ 179.9)$, $\mathcal{B} = \{\mathbf{x} \in \mathbb{N}^3 : \mathbf{p} \cdot \mathbf{x} \leq 500\}$. **True or false?**

1.2 — Plush symbols practice **Together**

$\mathbf{p} = (179.9,\ 79.9,\ 179.9)$, $\mathcal{B} = \{\mathbf{x} \in \mathbb{N}^3 : \mathbf{p} \cdot \mathbf{x} \leq 500\}$. **True or false?**

1. $[\in]$ $(0,0,3) \in \mathcal{B}$

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6. $[\Rightarrow]$ $(0,0,3) \in \mathcal{B}$ **if** $(1,1,1) \in \mathcal{B}$ **False** ($(1,1,1) \in \mathcal{B}$ but $539.7 > 500$)

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6. [$\Rightarrow$] $(0,0,3) \in \mathcal{B}$ **if** $(1,1,1) \in \mathcal{B}$ **False** ($(1,1,1) \in \mathcal{B}$ but $539.7 > 500$)
7. [contrapositive] If $\mathbf{p} \cdot (1,1,1) + \mathbf{p} \cdot (0,1,0) > 500$, then $(1,1,1) \notin \mathcal{B}$ **or** $(0,1,0) \notin \mathcal{B}$

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6. $[\Rightarrow]$ $(0,0,3) \in \mathcal{B}$ **if** $(1,1,1) \in \mathcal{B}$ **False** ($(1,1,1) \in \mathcal{B}$ but $539.7 > 500$)

7. [contrapositive] If $\mathbf{p} \cdot (1,1,1) + \mathbf{p} \cdot (0,1,0) > 500$, then $(1,1,1) \notin \mathcal{B}$ **or** $(0,1,0) \notin \mathcal{B}$ **False** ($519.6 > 500$, yet each basket alone is in $\mathcal{B}$)

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   each affordable $\not\Rightarrow$ jointly affordable

1.3 Sets and operations — names and definitions

Symbol	Name	Meaning
$\in$, $\notin$	element / not in	x belongs (or not) to A
$\subset$, $=$	subset, equality	$A \subset B$; $A = B$ iff mutual $\subset$
$\emptyset$, $\{\,\}$, $\{x \in S : \ldots\}$	sets	empty, roster, set-builder
$A \cup B$, $A \cap B$	union, intersection	in A or B; in both
A^c, $A \setminus B$	complement, difference	not in A; in A not B
$A \times B$	Cartesian product	pairs (a, b), $a \in A$, $b \in B$

In economics: budget set $\mathcal{B} \subset \mathbb{N}^3$; $(8.50, 2) \in \mathbb{Q} \times \mathbb{N}$.

$\mathcal{B} = \{\mathbf{x} \in \mathbb{N}^3 : \mathbf{p} \cdot \mathbf{x} \leq 500\}$, $\quad \mathcal{H} = \{\mathbf{x} \in \mathbb{N}^3 : \text{middle entry} \geq 1\}$ (at least one hedgehog).

Describe the set.

$\mathcal{B} = \{\mathbf{x} \in \mathbb{N}^3 : \mathbf{p} \cdot \mathbf{x} \leq 500\}$, $\quad \mathcal{H} = \{\mathbf{x} \in \mathbb{N}^3 : \text{middle entry} \geq 1\}$ (at least one hedgehog).

Describe the set.

1. $\mathcal{B} \cap \mathcal{H}$

$\mathcal{B} = \{\mathbf{x} \in \mathbb{N}^3 : \mathbf{p} \cdot \mathbf{x} \leq 500\}$, $\mathcal{H} = \{\mathbf{x} \in \mathbb{N}^3 : \text{middle entry} \geq 1\}$ (at least one hedgehog).

Describe the set.

1. $\mathcal{B} \cap \mathcal{H}$ affordable baskets with ≥ 1 hedgehog (e.g. $(0, 1, 0)$)

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2. $\mathcal{B} \cup \mathcal{H}$

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Describe the set.

1. $\mathcal{B} \cap \mathcal{H}$ affordable baskets with ≥ 1 hedgehog (e.g. $(0, 1, 0)$)
2. $\mathcal{B} \cup \mathcal{H}$ affordable baskets **or** baskets with ≥ 1 hedgehog

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2. $\mathcal{B} \cup \mathcal{H}$ affordable baskets **or** baskets with ≥ 1 hedgehog
3. $\mathcal{B} \setminus \mathcal{H}$

1.3 — Set operations (IKEA) Together

$\mathcal{B} = \{\mathbf{x} \in \mathbb{N}^3 : \mathbf{p} \cdot \mathbf{x} \leq 500\}$, $\mathcal{H} = \{\mathbf{x} \in \mathbb{N}^3 : \text{middle entry} \geq 1\}$ (at least one hedgehog).

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4. $(179.9, 79.9) \in \mathbb{Q} \times \mathbb{Q}$ but $(179.9, 79.9) \notin \mathcal{B}$

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5. $(1, 0, 0) \subset \mathcal{B}$ **False** $(1, 0, 0)$ is a basket: use $(1, 0, 0) \in \mathcal{B}$ (True); but $\{(1, 0, 0)\} \subset \mathcal{B}$ is True

1.4 Functions — names and definitions

Term	Notation	Meaning
Function	$f : A \to B$	each $x \in A$ maps to **exactly one** $f(x) \in B$
Domain	A	allowed inputs
Codomain	B	target set — every output must lie in B
Range	$f(A)$	outputs hit: $\{f(x) : x \in A\} \subseteq B$
Injective (one-to-one)		$f(x_1) = f(x_2) \Rightarrow x_1 = x_2$
Surjective (onto)		every $y \in B$ equals some $f(x)$
Operations	$(f \pm g)(x)$, $(cf)(x)$, $(fg)(x)$	$(f \pm g)(x) = f(x) \pm g(x)$; $(cf)(x) = c\,f(x)$; $(fg)(x) = f(x)g(x)$
Composition	$(f \circ g)(x)$	apply g first: $(f \circ g)(x) = f(g(x))$

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7. If $f(25) = \{22, 80\}$, is f a function? **No — a correspondence** (a function gives **exactly one** output per input; it's immoral to have a correspondence ☺)